

HIV TREATMENT AND MEDICAL SERVICES SYSTEM

Software Requirement Document

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I. Overview

1. Introduction

The **HIV Treatment and Medical Services System** is a web-based platform designed to streamline the management of HIV-related healthcare services, replacing inefficient manual and paper-based processes. The system aims to enhance patient care, improve operational efficiency, and ensure compliance with healthcare privacy regulations (e.g., HIPAA or equivalent local standards). It supports multiple user roles, secure data access, and integration with VNPay for payment processing, with plans for future scalability to connect with external health record systems and regional health databases.

- + Objectives
 - **Inefficient Appointment Scheduling:** Eliminates delays and errors in manual booking by allowing patients to schedule consultations and tests based on availability, reducing wait times.
 - **Fragmented Test Management:** Centralizes laboratory test management (CD4, viral load, screening, confirmatory tests) for seamless coordination between patients, doctors, and lab staff.
 - **Data Accessibility and Security:** Provides role-based access to patient data, ensuring privacy and compliance with regulations.
 - **Operational Efficiency:** Automates workflows for registration, consultations, and testing, reducing administrative burdens.
 - **Scalability:** Supports growing patient volumes with future integrations for external systems.
 - **Patient Engagement:** Empowers patients with access to medical records, appointment details, and test results.

- + Users
 - Guests: Browse general information about services.
 - Patients: Schedule appointments, make payments, and view medical records/test results.
 - Doctors: Reviews patient data, prescribes treatment and orders tests, examinations
 - Lab Staff: Process and record laboratory test results.
 - Registration Staff: Manage registrations, verify payments.
 - Managers: Oversee system performance, staff, patients, and financial transactions.

- + Constraints and Assumptions
 - Environment & Platform:
 - Backend: Node.js
 - Frontend: React
 - Hosting: Secure cloud platform for scalability and storage
 - Cross-browser compatibility: Chrome, Firefox, Safari
 - Responsive design for mobile and desktop access
 - Security:
 - Compliance with healthcare data privacy regulations (e.g., HIPAA)
 - JWT-based authentication with email verification
 - HTTPS for all data transmissions
 - Role-based access control (RBAC)
 - Integration:
 - VNPay for secure payment processing

- Assumptions:
 - Users have basic digital literacy for web navigation.
 - Internet access is available for all users.
 - VNPay API is reliable and available for integration.
- Constraints:
 - The system must handle high patient volumes without performance degradation.
 - Initial release focuses on core features; telehealth and external system integrations are planned for future phases.

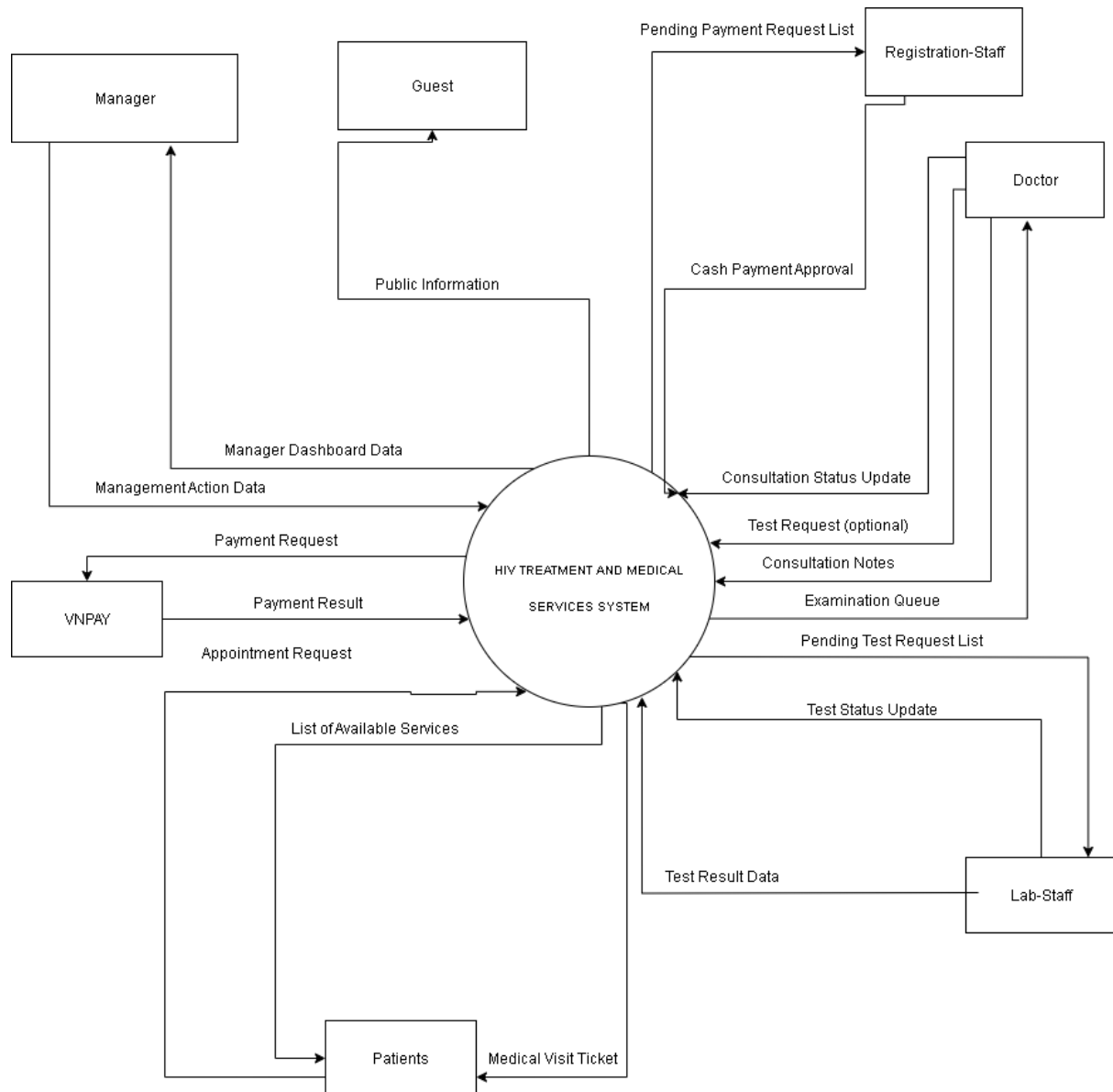


Diagram Description: The HIV Treatment and Medical Services System interacts with:

- Patients: Book appointments, make payments, and view records/results.
- Doctors: Access patient data, conduct consultations, and order tests.
- Lab Staff: Process test requests and enter results.
- Registration Staff: Verify payments and confirm appointments.
- Managers: Monitor system performance and manage users.
- VNPay: Processes payments for services.

2. Existing Systems

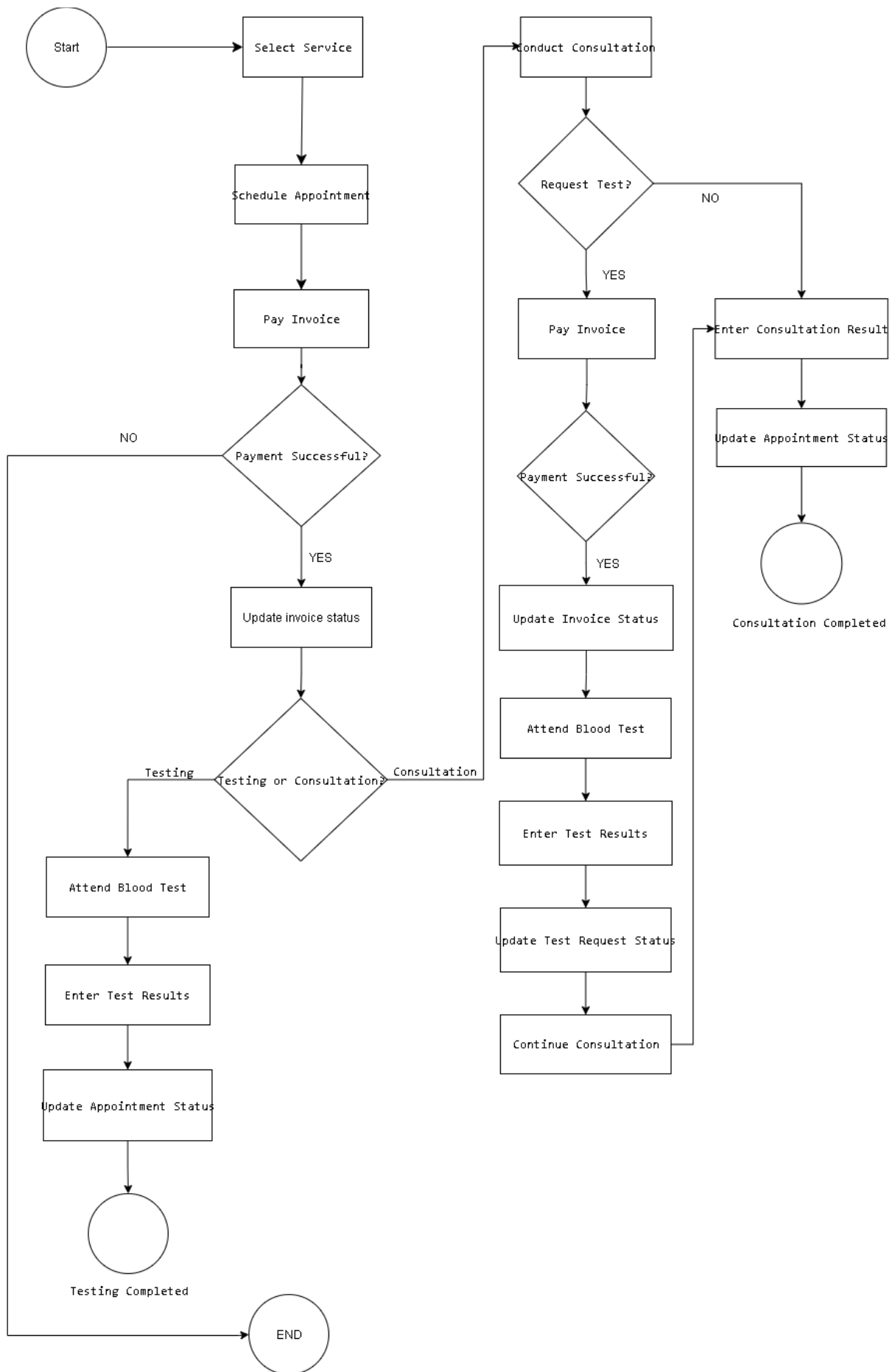
2.1 Galant Clinic

- Link: <https://galantclinic.com/hiv-aids/tu-van-hiv>
- Description: Provides HIV/AIDS consultation services with experienced specialists, focusing on testing, counseling, and treatment.
- Actors: Patients, Doctors
- Features:
 - Informational content about HIV testing and counseling
 - Consultation booking via contact forms
 - Emphasis on patient privacy Pros: Specialized in HIV/AIDS with credible expertise
 - Clear, patient-centric information
 - Strong focus on confidentiality
- Cons:
 - Lacks online appointment scheduling and payment integration
 - No support for multiple user roles (e.g., lab staff, managers)
 - Static, non-interactive interface
- Reference Value: Informs patient-facing informational content and privacy protocols.

2.2 IVIE - Bác Sĩ Ở

- Link: <https://ivie.vn/>
- Description: A telehealth platform offering online consultations, appointment booking, and health record management.
- Actors: Patients, Doctors, Support Staff
- Features:
 - Video call consultations
 - Appointment booking with priority scheduling Online medicine purchasing 24/7 support hotline
- Pros:
 - Modern, user-friendly interface
 - Telehealth capabilities for remote consultations
 - Integration with major hospitals (e.g., Vinmec)
- Cons:
 - Generalized focus, not specific to HIV/AIDS
 - Limited role support for lab staff or managers
 - Unclear payment gateway integration
 - Reference Value: Inspires patient-facing interface and potential telehealth features.

3. Business Process



Common Process		
#	Process Step	Description
1	Start	Patient initiates the process by accessing the system.
2	Select Service	Patient chooses between Testing or Consultation services.
3	Schedule Appointment	Patient books an appointment for either Testing or Consultation.
4	Pay Invoice	Patient makes payment for the selected service.
5	Payment Successful?	System checks if the payment is successful.
6	Update Invoice Status	System marks the invoice as "Paid" if payment is successful.
7	Testing or Consultation?	System determines whether the appointment is for Testing or Consultation.

Testing Branch		
#	Process Step	Description
9	Enter Test Results	Lab Staff inputs test results into the system.
10	Update Appointment Status	System marks the appointment status as "Testing Completed".
11	Testing Completed	Testing process is complete (end of Testing branch).

Consultation Branch		
#	Process Step	Description
9	Request Test?	Doctor decides whether to request an additional test (optional).
10	Pay Invoice (Test)	Patient makes payment for the additional test (if requested).
11	Payment Successful? (Test)	System checks if the payment for the test is successful.
12	Update Invoice Status (Test)	System marks the test invoice as "Paid" if payment is successful.
13	Attend Blood Test (Test Request)	Patient visits the facility for the requested blood test.
14	Enter Test Results (Test Request)	Lab Staff inputs test results into the system for the requested test.
15	Update Test Request Status	System marks the test request as "Completed".
#	Process Step	Description

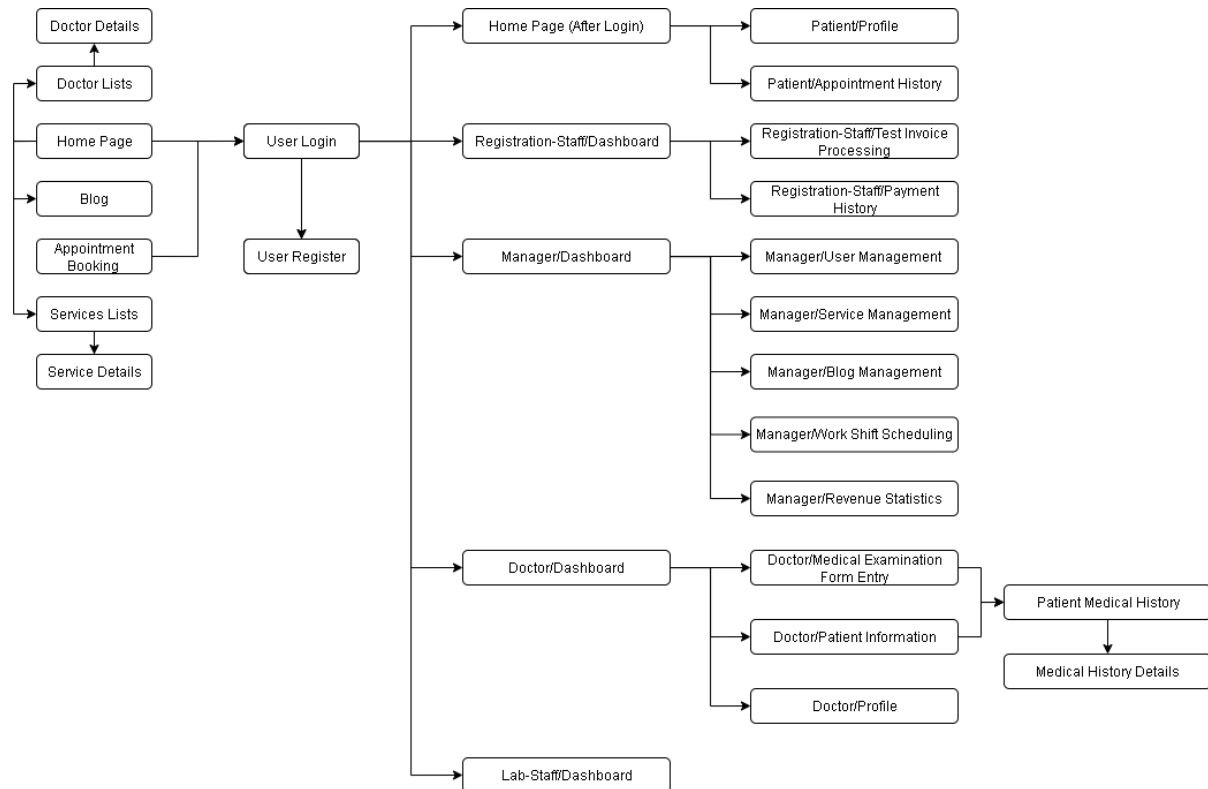
Consultation Branch		
17	Enter Consultation Results	Doctor inputs consultation results into the system (if no test or after test).
18	Update Appointment Status	System marks the appointment status for as " Completed".
19	Consultation Completed	Consultation process is complete (end of Consultation branch).

Failure Branch		
#	Process Step	Description
5.5	End	Process stops due to payment failure or completion.

4. System Functions

a. Screen Flow

b. Screen Details



#	Feature	Screen	Description
1	Authentication	User Login / User Register	Allows users to sign in to their existing accounts or create a new one.
2	Guest Features	Home Page	The main landing page for all users, providing an overview of the clinic and services.
3		Doctor Lists	Displays a list of all doctors, their specialties, and profiles.
4		Doctor Details	Displays detailed information about a doctor, including specialties, experience, work schedule, and related services. Allows users to book appointments directly with the doctor.
5		Services Lists	Displays a list of all medical services offered by the clinic.
6		Services Details	Displays detailed information about a medical service, including description, pricing, and associated doctors. Allows users to book appointments for the service.
7		Blog	Shows articles and news related to HIV treatment and health.
8		Appointment Booking	A form for guests or patients to book an appointment with a doctor or for a service.
9	Patient Features	Home page (after login)	The main dashboard for patients after logging in.
10		Patient Profile	Allows patients to view and update their personal and medical information.
11		Appointment History	Displays a list of the patient's past and upcoming appointments.

12	Doctor Features	Doctor Dashboard	The main dashboard for doctors, showing their schedule and tasks.
13		Medical Examination	A form for doctors to input clinical notes, diagnoses, and prescriptions during an exam.
14		Patient Information	Allows doctors to view detailed information about their patients, including current treatment plans (if any) and test results (if available).
15		Doctor Profile	Allows doctors to view and manage their own professional profile.
16		Patient Medical History	Displays a list of the patient's appointments, including details such as appointment date, time, doctor name, appointment ID, and creation timestamp.
17		Medical History Details	Provides detailed information about a specific medical record, including clinical notes, prescriptions, and associated test results.
18	Manager Features	Manager Dashboard	The main dashboard for managers with access to all management functions.
19		User Management	Allows managers to add, edit, and manage all user accounts in the system.
20		Service Management	Allows managers to add, edit, and manage medical services and their pricing.
21		Blog Management	Allows managers to create, edit, and publish blog posts.
22		Work Shift Scheduling	A tool for managers to create and manage doctor and staff work schedules.
23		Revenue Statistics	Displays reports and statistics on the clinic's revenue.
24	Registration Staff	Registration Staff Dashboard	The main dashboard for registration staff.
25		Test Invoice Processing	Allows staff to create and manage invoices for tests and services.
26		Payment History	Allows staff to view and track payment history.
27	Lab Staff	Lab Staff Dashboard	The main dashboard for lab staff, providing access to test request management and test result entry. Lab staff can view pending test requests, process them, and submit test results.

c. User Authorization

Screen	Patient	Doctor	Manager	Registration Staff	Lab Staff
Login/Register	X	X	X	X	X
Home Page	X	X	X	X	X
Doctor Lists	X	X	X	X	X
Services Lists	X	X	X	X	X
Blog	X	X	X	X	X
Appointment Booking	X				
Submit Appointment	X				
Select Doctor	X				
Select Service	X				
Home page (after login)	X				

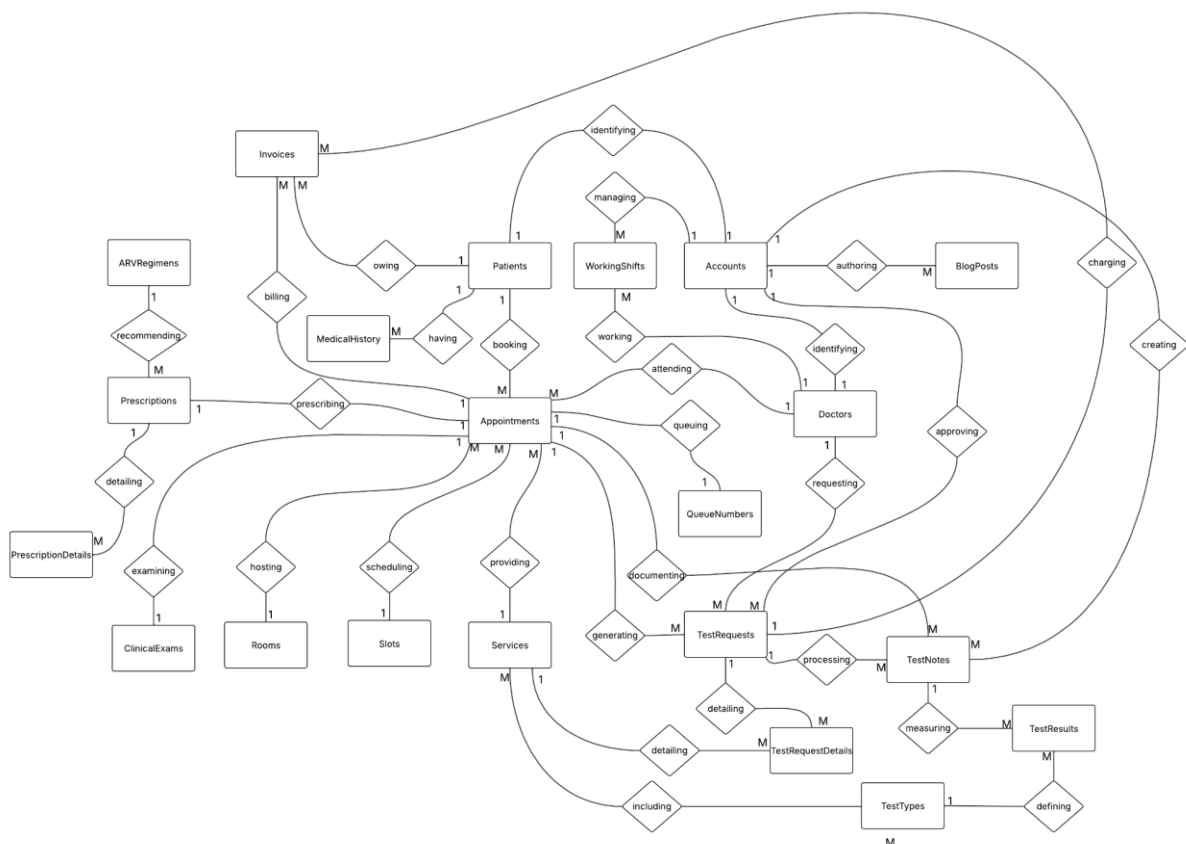
Patient Profile	X				
View Profile	X				
Update Profile	X				
Appointment History	X				
View Own Appointments	X				
Cancel Appointment	X				
Doctor Dashboard		X			
Medical Examination		X			
Add Clinical Notes		X			
Create Prescription		X			
Request Tests		X			
Patient Information		X			
View Patient Records		X			
View Medical History		X			
Doctor Profile		X			
View Own Profile		X			
Update Own Profile		X			
Manager Dashboard			X		
User Management			X		
Add User			X		
Update User			X		
Delete User			X		
View All Users			X		
Service Management			X		
Add Service			X		
Update Service			X		
Delete Service			X		
Blog Management			X		
Create Blog Post			X		
Edit Blog Post			X		
Delete Blog Post			X		
Work Shift Scheduling			X		
Create Schedule			X		
Update Schedule			X		
Revenue Statistics			X		
View Revenue Reports			X		
Registration Staff Dashboard				X	
Test Invoice Processing				X	
Create Invoice				X	
Process Payment				X	
Payment History				X	
View Payment Records				X	
Lab Staff Dashboard					X
Test Request Management					X
View Test Requests					X
Process Test Requests					X
Test Result Entry					X

Enter Test Results					X
Submit Test Results					X

In which:

- **Patient:** A user who can book appointments, view their medical records, and access test results.
- **Doctor:** A medical professional who can view patient information, manage medical records, and handle assigned appointments.
- **Manager:** A user responsible for managing system users, services, blogs, schedules, and viewing revenue statistics.
- **Registration Staff:** A user who handles appointment bookings, payment processing, and test invoice management.
- **Lab Staff:** A user who processes test requests and enters test results.

5. Entity Relationship Diagram



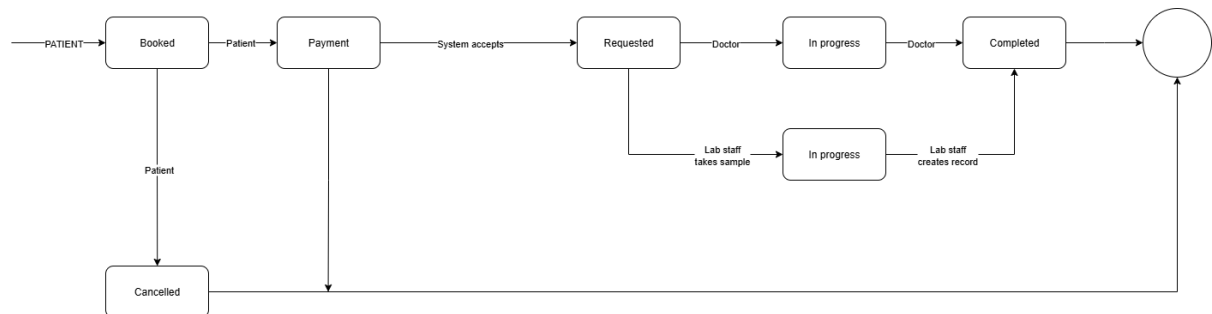
Entities Description

#	Entity	Description
1	Accounts	Manage login information and user roles in the system.
2	Patients	Personal and contact information of patients.
3	Doctors	Detailed doctor profiles, linked to their user accounts.
4	ARVRegimens	List of ARV treatment regimens by patient groups.
5	Rooms	List of rooms used for examination and lab tests.
6	Slots	Fixed time slots used for scheduling appointments or tests.

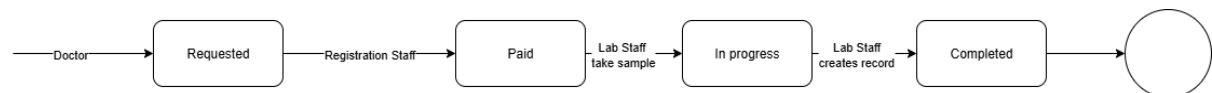
7	Services	Medical services provided in the system such as exams, consultations, tests.
8	TestTypes	Types of lab tests, measurement units , and result formats.
9	ServicesTestTypes	Many-to-many relationship between services and test types.
10	Appointments	Appointments between patients and doctors or test rooms.
11	TestRequests	Test request issued by for patients.
12	TestRequestDetails	Detailed services within a test request.
13	WorkingShifts	Daily work shifts of doctors and medical staff.
14	TestNotes	Notes and summaries recorded during lab tests.
15	TestResults	Results for each test type within a test note.
16	MedicalHistory	Patient history of HIV treatment and related medical conditions.
17	ClinicalExams	Clinical exam results recorded during appointments.
18	Prescriptions	Medical prescriptions issued after consultations.
19	PrescriptionDetails	Detailed list of drugs prescribed in each prescription.
20	BlogPosts	Blog posts created by medical staff within the system.
21	Invoices	Billing records for appointments and test services.
22	QueueNumbers	Queue management for consultations and tests on a daily basis.

6. State transition diagram

APPOINTMENTS



TEST REQUEST



II. Functional Requirements

1. User Authentication and Authorization

a. Patient Login

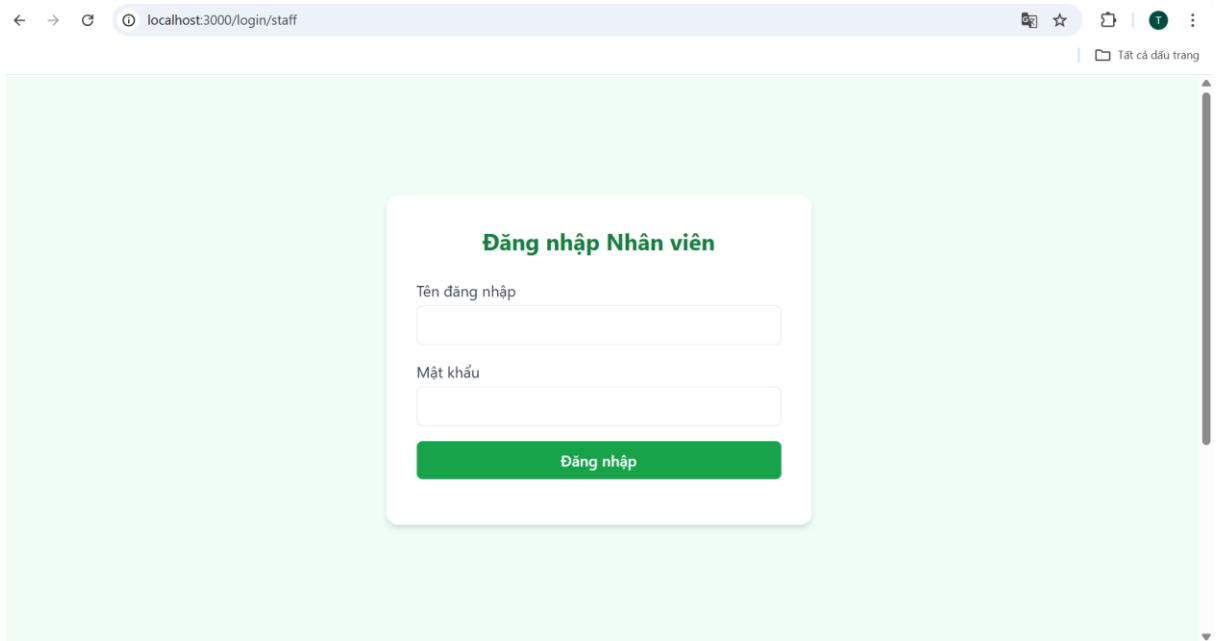
- Function Trigger: User accesses the path "/login/patient"
- Function Description:
 - Actors: Patient
 - Purpose: Allows patients to log into the system to access personal services.
 - Interface: Login form with username and password.
 - Data Processing: Verify login information, generate authentication token.
- Screen Layout:

The screenshot shows a web browser window with the address bar displaying "localhost:3000/login/patient". The page header includes the "H+ HIV Care Center" logo and navigation links: "Giới thiệu", "Chuyên Gia - Bác sĩ", "Tin tức", "Dịch vụ HIV", "Đặt lịch", "Đăng nhập", and "Đăng ký". The main content area features a light green background with a white login form titled "Đăng nhập Bệnh nhân". The form contains two input fields: "Tên đăng nhập" (Username) and "Mật khẩu" (Password). Below the fields is a green "Đăng nhập" button. Underneath the button, there is a link "Chưa có tài khoản? Đăng ký ngay" and a blue link "Quên mật khẩu?".

- Function Details:
 - The user enters login information (username and password).
 - The system checks the validity of the information.
 - If valid, the system generates an authentication token and redirects the user to the patient home page.
 - If invalid, an error message is displayed.
 - There is a "Forgot Password" function to recover the account.

b. Staff Login

- Function Trigger: User accesses the path "/login/staff"
- Function Description:
 - Actors: Staff (Doctor, Laboratory Staff, Registration Staff, Manager)
 - Purpose: Allows staff to log in to the system to access staff functions
 - Interface: Login form with username and password
 - Data Processing: Verify login information, check roles, generate authentication tokens
- Screen Layout:



- Function Details:
 - The employee enters the username and password.
 - The system checks the validity of the information and determines the employee's role.
 - Based on the role, the system redirects to the corresponding dashboard.
 - If the information is invalid, an error message is displayed.

c. Patient Registration

- Function Trigger: User accesses the path "/register"
- Function Description:
 - Actors: New Patients
 - Purpose: Allow new patients to register an account
 - Interface: Registration form with personal information
 - Data Processing: Create new account, save patient information to database
- Screen Layout:

Đăng ký tài khoản

Tạo tài khoản để sử dụng dịch vụ y tế

Thông tin đăng nhập

Tên đăng nhập *

Mật khẩu *

Mật khẩu phải có ít nhất 8 ký tự, bao gồm chữ thường, chữ hoa, số và ký tự đặc biệt (@!%*?&)

Xác nhận mật khẩu *

Thông tin cá nhân

Họ và tên *

Ngày sinh *



Giới tính *



Email *

Số điện thoại *

Địa chỉ *

Mã OTP *

Lấy
OTP

Đăng ký tài khoản

- Function Details:
 - User enters personal information (full name, date of birth, gender, address, email, phone number, password)
 - The system checks the validity of the information (email format, strong password, valid phone number)

- Send confirmation code to email to verify identity
- If verification is successful, create an account and redirect to the login page

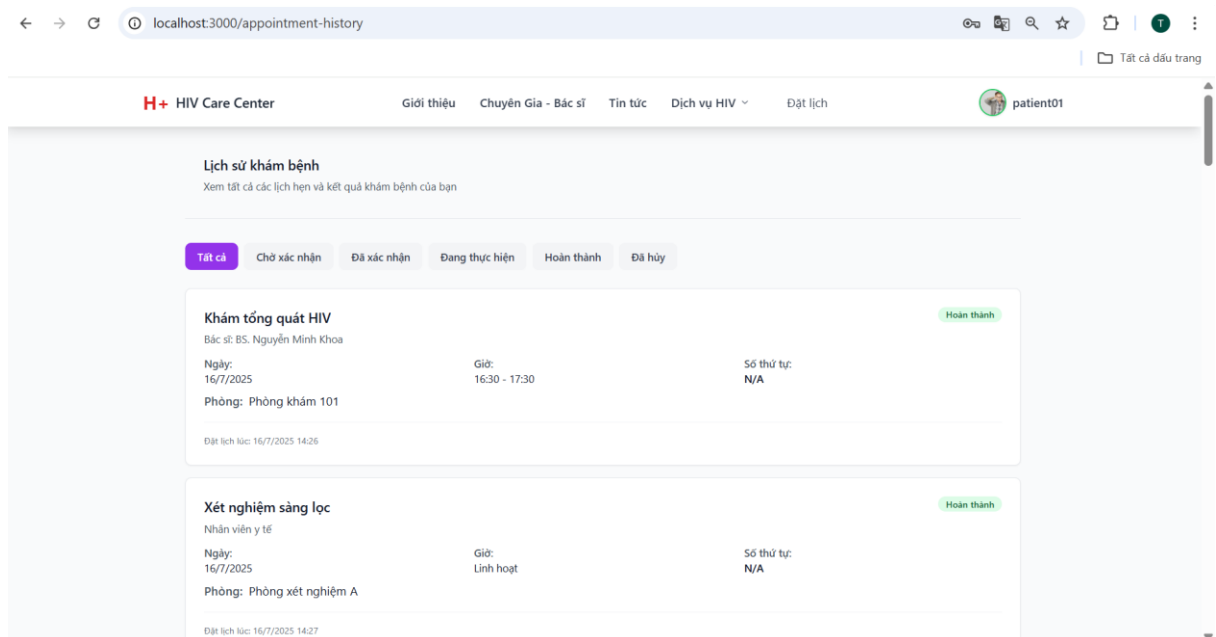
2. Patient Management

a. Patient Profile Management

- Function Trigger: Logged in patients access the path `"/profile"`
- Function Description:
 - Actors: Patients
 - Purpose: Allows patients to view and update personal information
 - Interface: Form to display and edit personal information
 - Data Processing: Retrieves and updates patient information in the database
- Screen Layout: Patient Profile Screen
- Function Details:
 - Display patient's personal information (full name, date of birth, gender, address, email, phone number)
 - Allow patient to update information (except certain fields such as date of birth)
 - Check validity of new information
 - Update information to database and display success message

b. Appointment History

- Function Trigger: Logged in patients access the path `"/appointment-history"`
- Function Description:
 - Actors: Patients
 - Purpose: Allows patients to view their appointment history
 - Interface: List of appointments with detailed information
 - Data Processing: Retrieves appointment history from database
- Screen Layout:



- Function Details:
 - Show a list of medical appointments by time (upcoming, completed)
 - Each appointment shows information: date, time, doctor, service, status

- Allows to filter appointments by status (pending confirmation, confirmed, completed, canceled)
- Show appointment details when clicking on a specific appointment

3. Appointment Booking System

a. Service Selection

- Function Trigger: User accesses the path "/service/:serviceId"
- Function Description:
 - Actors: Patients
 - Purpose: Allows patients to view detailed information about medical services
 - Interface: Page displaying service information with "Book Appointment" button
 - Data Processing: Retrieves service information from database
- Screen Layout:

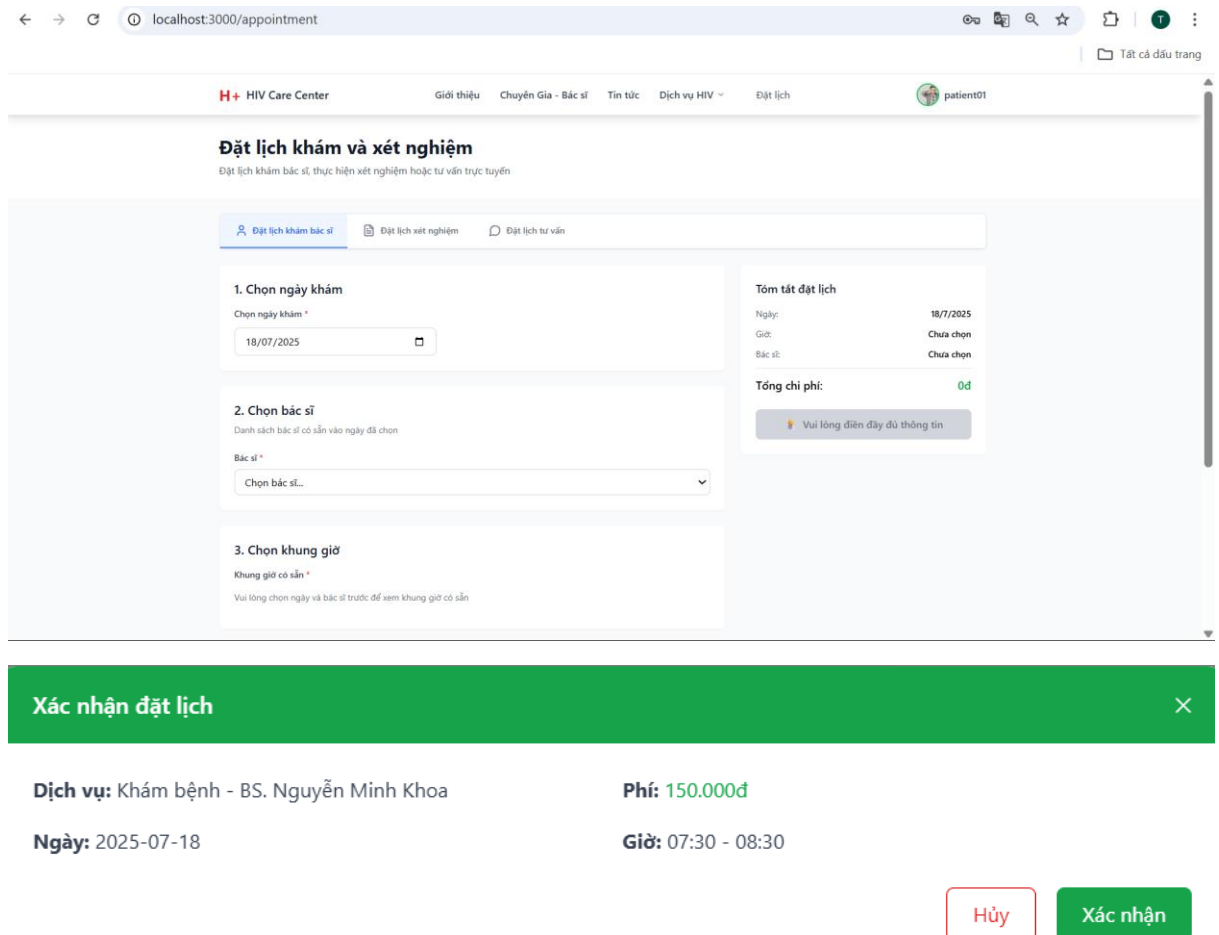
The screenshot shows a web interface for the HIV Care Center. At the top, there is a navigation bar with the logo 'H+ HIV Care Center' and links for 'Giới thiệu', 'Chuyên Gia - Bác sĩ', 'Tin tức', 'Dịch vụ HIV', and 'Đặt lịch'. A user profile icon labeled 'patient01' is in the top right corner. Below the navigation bar, a green button labeled 'Quay lại trang chủ' is visible. The main content area features a large white card titled 'Xét nghiệm CD4 và Viral Load' with the subtitle 'Đếm tế bào CD4 và đo tải lượng virus HIV'. This card contains two green buttons: '550.000đ' (labeled 'Chi phí') and 'Xét nghiệm' (labeled 'Loại dịch vụ'). Below this card, there are two more white cards. The left card, titled 'Đặc điểm nổi bật', lists five features with green checkmarks: 'Xét nghiệm chính xác và đáng tin cậy', 'Thực hiện bởi đội ngũ chuyên môn cao', 'Bảo mật thông tin tuyệt đối', 'Kết quả nhanh chóng', and 'Hỗ trợ tư vấn chuyên nghiệp'. The right card, titled 'Đặt lịch dịch vụ', contains a 'Đặt lịch ngay' button, the service name 'Xét nghiệm CD4 và Viral Load', the price 'Giá dịch vụ 550000', a date selection field labeled 'Chọn ngày khám' with a calendar icon, and another 'Đặt lịch ngay' button. At the bottom left, a third white card titled 'Quy trình thực hiện' shows a three-step process: 1. Đăng ký và khai báo y tế, 2. Lấy mẫu theo yêu cầu, and 3. Thực hiện xét nghiệm.

- Function Details:
 - Show detailed information about the medical service (name, description, price, time of implementation, necessary preparation)
 - Show "Book an appointment" button to go to the appointment booking page

b. Appointment Booking

- Function Trigger: User accesses the "/appointment" path or from the service details page.

- **Function Description:**
 - Actors: Patients
 - Purpose: Allows patients to book medical appointments
 - Interface: Appointment form with service selection, doctor, date and time
 - Data Processing: Checks for available appointments, creates new appointments in the database
- **Screen Layout:**



Đặt lịch khám và xét nghiệm
Đặt lịch khám bác sĩ, thực hiện xét nghiệm hoặc tư vấn trực tuyến

1. Chọn ngày khám
Chọn ngày khám *
18/07/2025

2. Chọn bác sĩ
Danh sách bác sĩ có sẵn vào ngày đã chọn
Bác sĩ *
Chọn bác sĩ...

3. Chọn khung giờ
Khung giờ có sẵn *
Vui lòng chọn ngày và bác sĩ trước để xem khung giờ có sẵn

Tóm tắt đặt lịch
Ngày: 18/7/2025
Giờ: Chưa chọn
Bác sĩ: Chưa chọn
Tổng chi phí: 0đ
Vui lòng điền đầy đủ thông tin

Xác nhận đặt lịch

Dịch vụ: Khám bệnh - BS. Nguyễn Minh Khoa
Phí: 150.000đ
Ngày: 2025-07-18
Giờ: 07:30 - 08:30

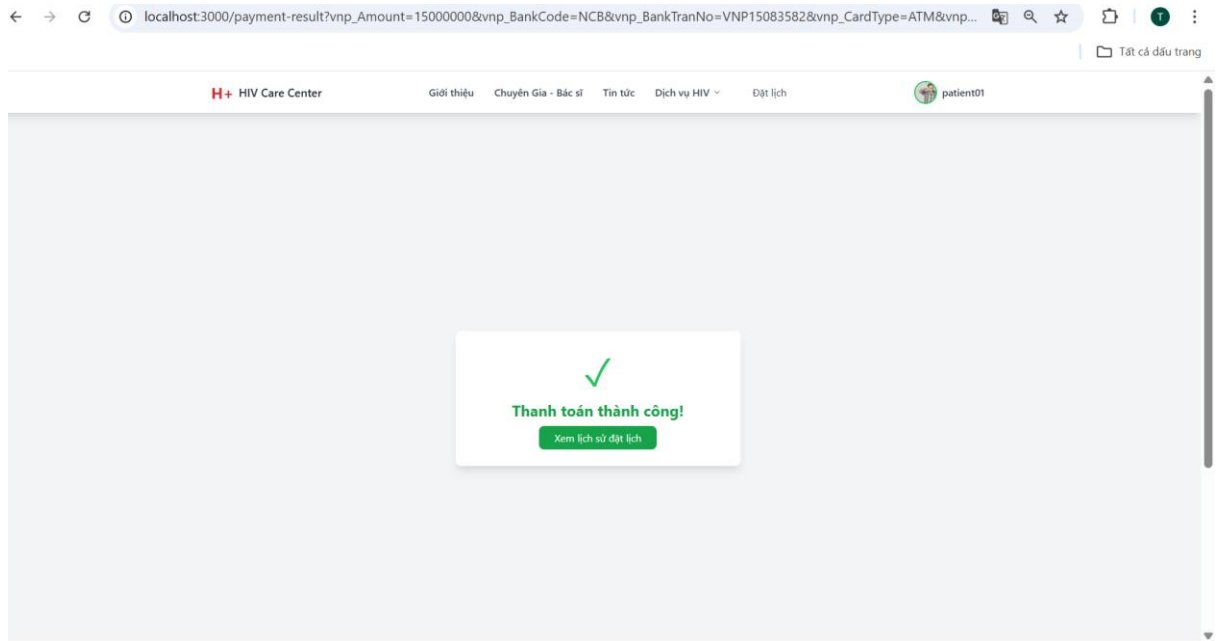
Hủy **Xác nhận**

- **Function Details:**
 - Patient selects desired medical service
 - System displays list of doctors matching selected service
 - Patient selects desired doctor and date and time from available slots
 - Patient confirms appointment
 - System checks for schedule conflicts and creates new appointment
 - Notification of successful appointment confirmation

c. Payment Processing

- **Function Trigger:** After successful booking
- **Function Description:**
 - Actors: Patients
 - Purpose: Allows patients to pay for services
 - Interface: Payment page with payment methods and order information
 - Data Processing: Process payments, update payment status in database

- Screen Layout:

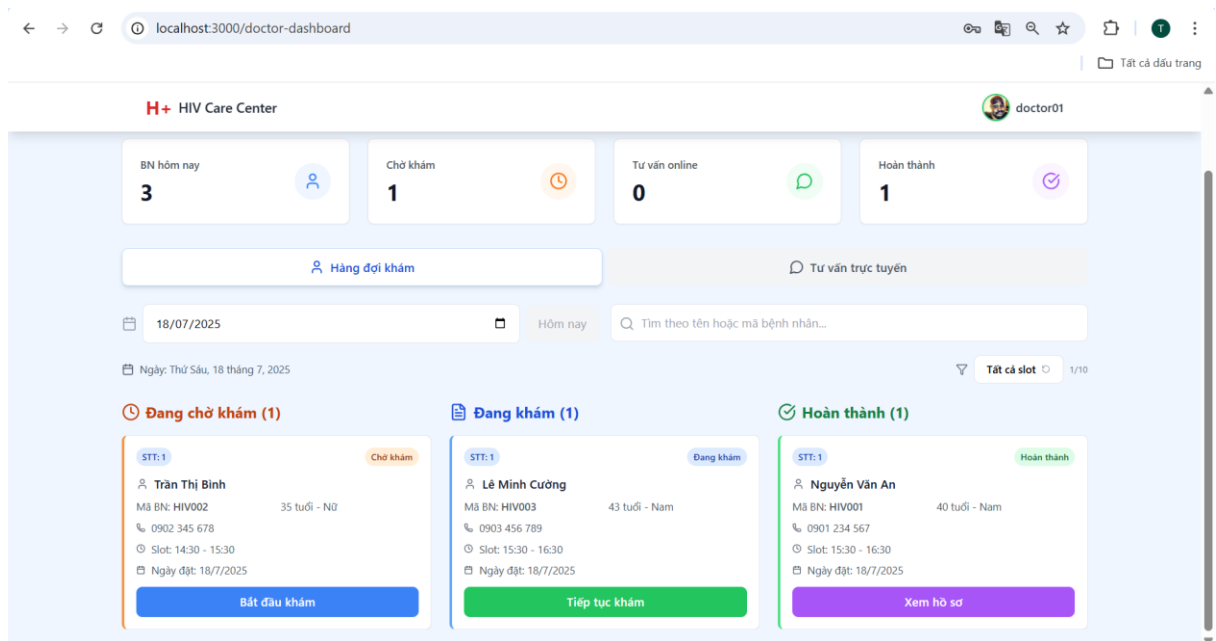


- Function Details:
 - Display order information (services, prices, taxes, totals)
 - Provide payment methods (credit cards, bank transfers, e-wallets)
 - Process payments via integrated payment gateway
 - Display payment results (success/failure)

4. Doctor Management System

a. Doctor Dashboard

- Function Trigger: The doctor has logged in and accesses the path "/doctor-dashboard"
- Function Description:
 - Actors: Doctors
 - Purpose: Shows an overview of the appointment schedule and patients for the day
 - Interface: Dashboard with statistics, patient list by status
 - Data Processing: Retrieves appointment and patient information from the database
- Screen Layout:



- Function Details:
 - Display number of patients by status (waiting, in progress, completed)
 - Display patient list by status
 - Allow doctors to search for patients by name or ID
 - Allow doctors to filter list by date
 - Display notification when new patients arrive

b. Patient Examination

- Function Trigger: Doctor selects patient from dashboard
- Function Description:
 - Actors: Doctor
 - Purpose: Allows doctors to examine patients and record results
 - Interface: Examination form with tabs (patient information, clinical examination, prescription)
 - Data Processing: Save examination results and prescriptions to database
- Screen Layout:

← Quay lại

Phiếu khám bệnh

Nhập thông tin khám lâm sàng

Thông tin bệnh nhân

Họ tên:
Lê Minh Cường

Tuổi/Giới:
43 tuổi - Nam

SDT:
0903456789

Mã BN:
HIV003

Giờ hẹn:
15:30 - 16:30

Địa chỉ:
789 Hai Bà Trưng, Q1, TP.HCM

Phác đồ ARV hiện tại

Tên phác đồ:
TDF/3TC/EFV

Loại thuốc:
Tenofovir 300mg + Lamivudine 300mg + Efavirenz 600mg

Ngày bắt đầu:
15-07-2025

Kết quả xét nghiệm gần nhất

Cập nhật lần cuối: 16:30:56

Cập nhật

Sàng lọc HIV:
Positive
2025-07-03 21:44:08
Tải lượng virus không phát hiện được

Khẳng định HIV:
Âm tính
2025-07-14 23:11:16
sfddf

Số lượng CD4:
600 cells/mm³
2025-07-15 17:57:44
9999

Tải lượng virus:
34 copies/ml
2025-07-15 17:57:44
9999

Chỉ định xét nghiệm

Thêm chỉ định

Chưa có chỉ định xét nghiệm

Nhấn "Thêm chỉ định" để tạo chỉ định xét nghiệm mới

Khám hiện tại

Lịch sử khám

Sinh hiệu

Nhịp tim

76

bpm

Huyết áp

120/80

mmHg

Nhiệt độ

37

°C

Cân nặng

78

kg

Chiều cao

178

cm

BMI

24.6

kg/m²

Bình thường

Dấu hiệu lâm sàng

Triệu chứng và dấu hiệu lâm sàng

Mệt mỏi, Chóng mặt

Dấu hiệu thường gặp (click để thêm nhanh)

+ Sốt

+ Ho khan

+ Ho có đờm

+ Khó thở

+ Mệt mỏi

+ Giảm cân

+ Phát ban

+ Tiêu chảy

+ Đau đầu

+ Buồn nôn

+ Hạch to

+ Đau khớp

+ Tụt cân

+ Chán ăn

+ Đau bụng

+ Táo bón

+ Chóng mặt

+ Mất ngủ

+ Lo âu

+ Trầm cảm

Chẩn đoán

* Chẩn đoán chính

Kết quả xét nghiệm khẳng định có sự hiện diện của virus HIV trong cơ thể.

Chẩn đoán phụ

Bệnh nhân hiện tại không có các bệnh lý đồng mắc nghiêm trọng liên quan đến HIV được ghi nhận.

Hướng dẫn chẩn đoán

- Chẩn đoán chính là bắt buộc để hoàn thành khám
- Sử dụng mã ICD-10 khi có thể
- Chẩn đoán phụ cho các bệnh lý kèm theo
- Ghi rõ mức độ nghiêm trọng và tiến triển

Kế hoạch điều trị

Lựa chọn điều trị

☐ Tiếp tục phác đồ hiện tại ☒ Thay đổi phác đồ điều trị

Chọn phác đồ ARV mới *

ABC/3TC/DTG (Người lớn)

Phác đồ ABC/3TC/DTG

Người lớn

Thuốc chính trong phác đồ:



Abacavir (ABC)

Chưa cấu hình



Lamivudine (3TC)

Chưa cấu hình



Dolutegravir (DTG)

Chưa cấu hình

Thuốc hỗ trợ



Vitamin B

✓ Đã cấu hình: 2 viên - 2 lần/ngày - 30 ngày

Thông tin thuốc hỗ trợ

Tên thuốc *

Nhập tên thuốc...

Liều dùng *

Chọn liều dùng...

Số lần/ngày *

Chọn số lần/ngày...

Số ngày dùng *

VD: 30

Cách dùng

VD: Uống sau ăn

Ghi chú

Ghi chú thêm về thuốc...

Hủy

Lưu

Lời khuyên và tư vấn *

Duy trì chế độ ăn uống cân bằng, đầy đủ dưỡng chất để tăng cường sức đề kháng. Hạn chế thực phẩm chế biến sẵn, nhiều đường và chất béo.

Kế hoạch tái khám *

Tái khám định kỳ mỗi 3 tháng để xét nghiệm tải lượng virus, CD4 và đánh giá lâm sàng.

Ghi chú của bác sĩ *

Nếu có bất kỳ triệu chứng bất thường hoặc khẩn cấp, vui lòng liên hệ ngay với chúng tôi qua số điện thoại: 0909090909

← Quay lại Dashboard

Lưu tạm

Hoàn thành khám

Lưu tạm: Lưu và quay lại, có thể thiếu thông tin Hoàn thành: Kết thúc khám, cần đầy đủ thông tin

- Function Details:
 - Display basic patient information
 - Allow the doctor to enter vital signs (blood pressure, heart rate, temperature, weight, height, BMI)
 - Allow the doctor to enter clinical symptoms and diagnosis
 - Allow the doctor to prescribe medication (ARV regimen, supportive medication)
 - Allow the doctor to order tests
 - Save examination results and change patient status to "completed"

c. Doctor Test Request

- Function Trigger: The doctor requested tests for the patient.
- Function Description:
 - Actors: Doctors
 - Purpose: Allows doctors to select the type of test for the patient
 - Interface: Form to display and edit information
 - Data Processing: Retrieve and update information for Registration Staff to review
- Screen Layout:

Chi định xét nghiệm + Thêm chỉ định

Chọn loại xét nghiệm

- ☒ **Xét nghiệm CD4 và Viral Load**
Đếm tế bào CD4 và đo tải lượng virus HIV
550.000 VNĐ
Ghi chú:
Nhập ghi chú cho xét nghiệm này...
- ☐ **Xét nghiệm khẳng định**
Xét nghiệm khẳng định HIV, CD4 và Viral Load
80.000 VNĐ
- ☐ **Xét nghiệm sàng lọc**
Sàng lọc HIV sơ bộ
120.000 VNĐ

Tạo chỉ định Hủy

Chi định xét nghiệm

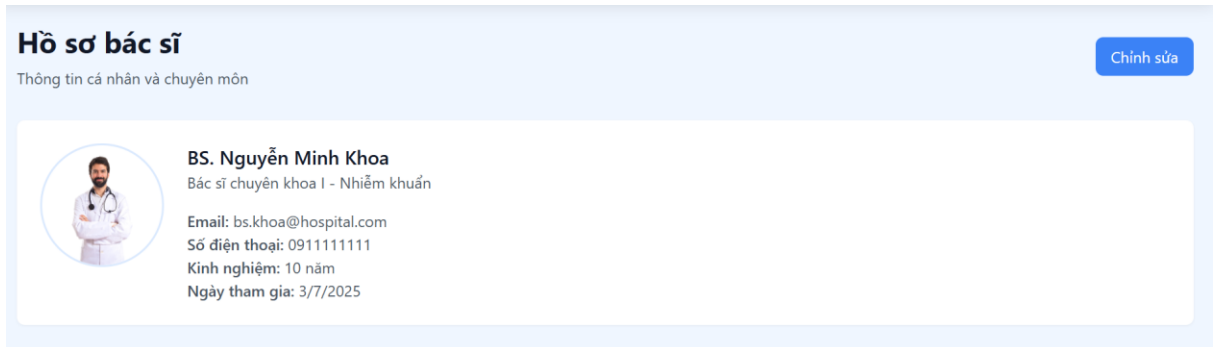
🕒 **Đang chờ xét nghiệm**
Ngày tạo: 2025-07-18 16:47
Các xét nghiệm đã chỉ định:
• Xét nghiệm CD4 và Viral Load

⚠️ Không thể tạo chỉ định mới. Bệnh nhân đang có chỉ định xét nghiệm chưa hoàn thành.

- Function Details:
 - Show the services that the doctor prescribes for testing
 - The doctor has the right to prescribe multiple services for the patient

d. Doctor Profile Management

- Function Trigger: The doctor has logged in and accessed the path "/profile"
- Function Description:
 - Actors: Doctors
 - Purpose: Allows doctors to view and update personal and professional information
 - Interface: Form to display and edit information
 - Data Processing: Retrieves and updates doctor information in the database
- Screen Layout:



Hồ sơ bác sĩ
Thông tin cá nhân và chuyên môn

BS. Nguyễn Minh Khoa
Bác sĩ chuyên khoa I - Nhiễm khuẩn

Email: bs.khoa@hospital.com
Số điện thoại: 0911111111
Kinh nghiệm: 10 năm
Ngày tham gia: 3/7/2025

Chỉnh sửa

- Function Details:
 - Display personal and professional information of the doctor
 - Allow the doctor to update information (email, phone number, qualifications, experience)
 - Display statistics on the number of patients examined, patient reviews

5. Lab Staff Management System

a. Lab Staff Dashboard

- Function Trigger: The lab staff logged in and accessed the path "/lab-staff"
- Function Description:
 - Actors: Testing staff
 - Purpose: Displays an overview of the tests to be performed during the day
 - Interface: Dashboard with statistics, test list by status
 - Data Processing: Retrieves test information from the database
- Screen Layout:

HIV Care Center

labstaff01

Xét nghiệm CD4 và Viral Load

2

Xét nghiệm sàng lọc

0

Xét nghiệm khẳng định

3

Hoàn thành

1

18/07/2025

Hôm nay

Tìm theo tên hoặc mã bệnh nhân...

Ngày: 18/7/2025

Chờ xử lý

2

#1 - Trần Thị Bình

Mã bệnh nhân: 002

Dịch vụ: Xét nghiệm khẳng định

Ngày đặt: 18/7/2025

Nguồn: Tự đăng ký

Chờ xử lý

#2 - Nguyễn Văn An

Mã bệnh nhân: 001

Dịch vụ: Xét nghiệm CD4 và Viral Load, Xét nghiệm khẳng định

Ngày đặt: 18/7/2025

Nguồn: Bác sĩ chỉ định

Chờ xử lý

Đang xử lý

1

#1 - Lê Minh Cường

Mã bệnh nhân: 003

Dịch vụ: Xét nghiệm CD4 và Viral Load

Ngày đặt: 18/7/2025

Nguồn: Bác sĩ chỉ định

Đang xử lý

Hoàn thành

1

#1 - Nguyễn Văn An

Mã bệnh nhân: 001

Dịch vụ: Xét nghiệm khẳng định

Ngày đặt: 18/7/2025

Nguồn: Tự đăng ký

Hoàn thành

- **Function Details:**
 - Display number of tests by type (CD4 and Viral Load, screening tests, confirmatory tests)
 - Display list of tests by status (requested, in progress, completed)
 - Allow staff to search for tests by patient name or ID Allow staff to filter list by date

b. Test Processing

- **Function Trigger:** Test staff selects test from dashboard
- **Function Description:**
 - **Actors:** Laboratory staff
 - **Purpose:** Allows laboratory staff to enter test results
 - **Interface:** Laboratory form with patient information and result entry fields
 - **Data Processing:** Saves test results to database
- **Screen Layout:**

Page | 28

Chi tiết xét nghiệm

×

Thông tin bệnh nhân

Họ tên: Lê Minh Cường

Mã bệnh nhân: 3

Tuổi: 43

Giới tính: male

Điện thoại: 0903456789

Email: lmc@email.com

Địa chỉ: 789 Hai Bà Trưng, Q.1, TP.HCM

Trạng thái: in_progress

Loại xét nghiệm

CD4

500

cells/mm³

HIV Viral Load

44

copies/ml

Ghi chú:

Nhập ghi chú (tùy chọn)

Hoàn thành

Đóng

- Function Details:
 - Display basic patient information and test type
 - Allow staff to update test status (in process, completed)
 - Allow staff to enter test results and notes Save test results and change status to "complete"
 - Automatically notify physician when test is completed

6. Registration Staff Management System

a. Registration Staff Dashboard

- Function Trigger: The logged in registration staff accesses the path "/registration-staff"
- Function Description:
 - Actors: Registration staff
 - Purpose: Display overview of registration and payment requests
 - Interface: Dashboard with statistics, request list by status
 - Data Processing: Retrieve registration and payment information from database

Page | 29

- Screen Layout:

H+ HIV Care Center regstaff01

Quản lý Xét nghiệm - Registration Staff

Xử lý đơn xét nghiệm và thu tiền từ bệnh nhân - Ngày 18/7/2025

Xử lý đơn XN Lịch sử thanh toán

Đơn chờ xử lý
1

Doanh thu hôm nay
550.000 đ
Tổng tiền thu được hôm nay

Đã xử lý hôm nay
1
Số đơn đã hoàn thành hôm nay

🔍 Tìm kiếm...

Danh sách đơn xét nghiệm chờ xử lý
Tổng cộng 1 đơn

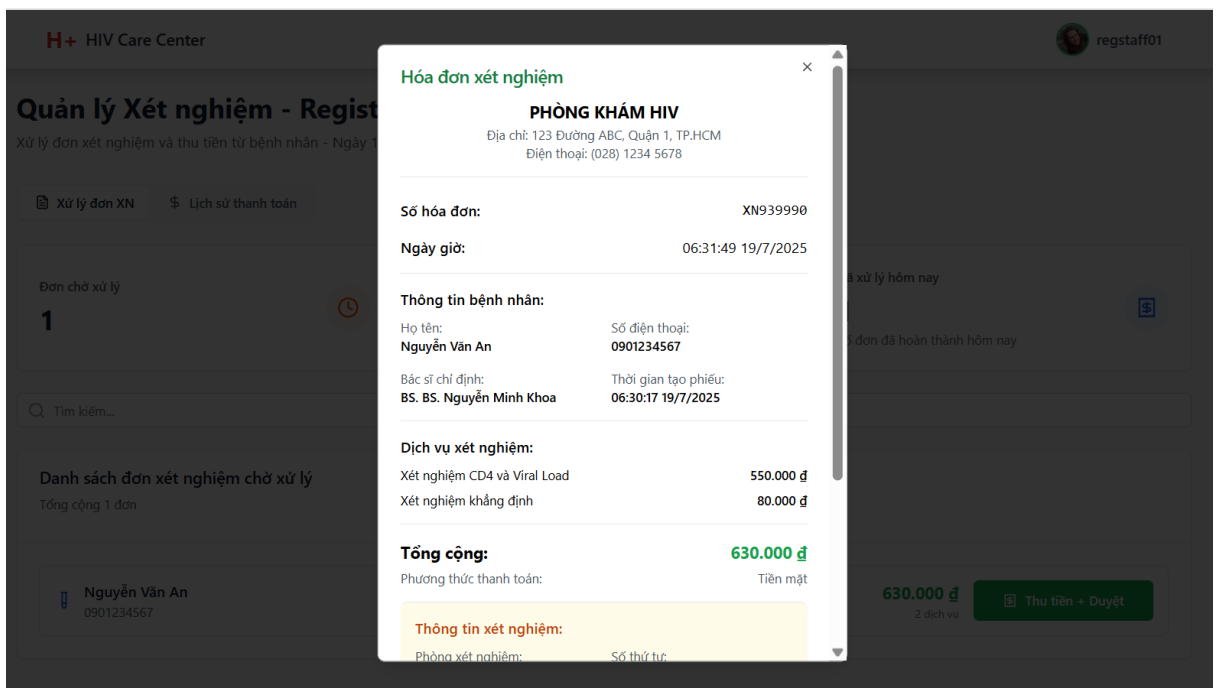
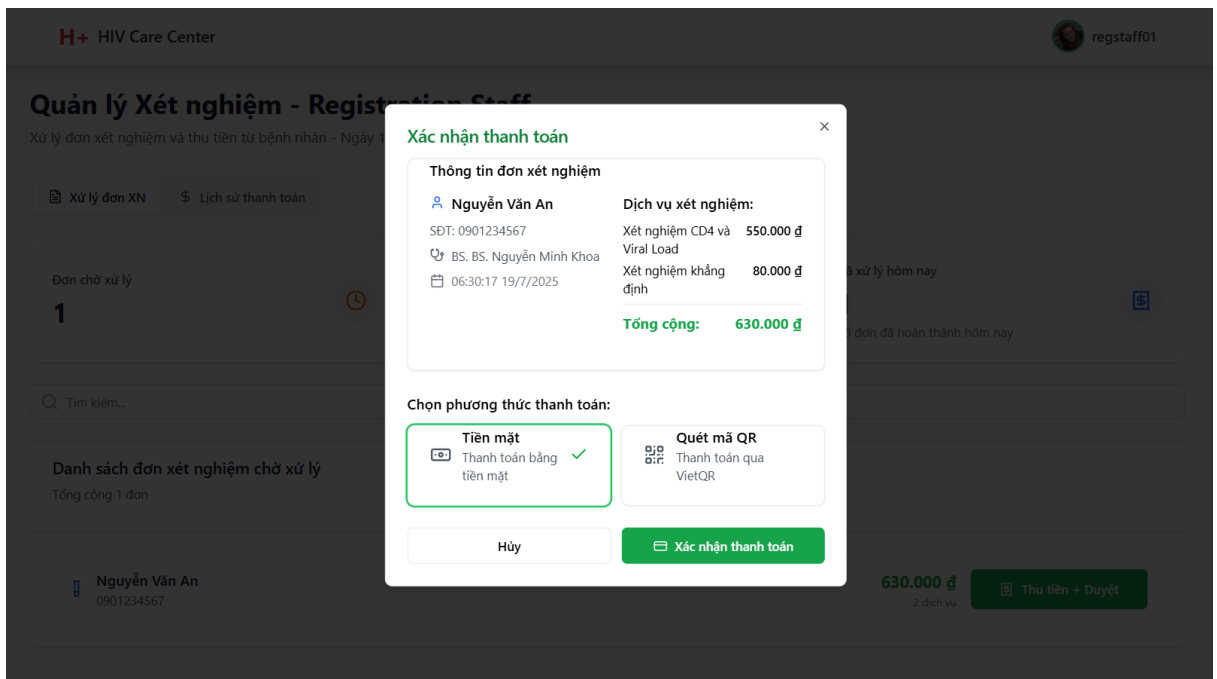
Nguyễn Văn An 0901234567	630.000 đ 2 dịch vụ	Thu tiền + Duyệt
------------------------------------	--	-------------------------

- Function Details:

- Display number of registration requests by status (pending confirmation, confirmed, completed)
- Display list of payment requests
- Allow staff to search requests by patient name or ID
- Allow staff to filter list by date

b. Payment Verification

- Function Trigger: The registration staff selects the payment request from the dashboard
- Function Description:
 - Actors: Registration staff
 - Purpose: Allow staff to confirm payment from patient
 - Interface: Payment confirmation form with detailed information
 - Data Processing: Update payment status in database
- Screen Layout:



- Function Details:
 - Display payment request details (service, price, patient)
 - Allow staff to confirm payment received
 - Allow staff to print invoices
 - Update payment status

c. Payment History

- Function Trigger: Staff click to view payment history
- Function Description:
 - Actors: Registered staff
 - Purpose: Allows staff to view payment history for the day
 - Interface: List of payment transactions for the day
 - Data Processing: Get payment list from database

- Screen Layout:

Quản lý Xét nghiệm - Registration Staff

Xử lý đơn xét nghiệm và thu tiền từ bệnh nhân - Ngày 18/7/2025

 Xử lý đơn XN

 Lịch sử thanh toán

 Tìm kiếm...

Lịch sử thanh toán

Tổng cộng 2 giao dịch



Nguyễn Văn An

SĐT: 0901234567

630.000 đ

Tiền mặt

Dịch vụ:

Xét nghiệm CD4 và Viral Load

550.000 đ

Xét nghiệm kháng định

80.000 đ

Thanh toán lúc:

23:31:49 18/7/2025

Trạng thái:

Đã thanh toán



Lê Minh Cường

SĐT: 0903456789

550.000 đ

Tiền mặt

Dịch vụ:

Xét nghiệm CD4 và Viral Load

550.000 đ

Thanh toán lúc:

23:28:35 18/7/2025

Trạng thái:

Đã thanh toán

Quản lý Xét nghiệm - Registration Staff

Xử lý đơn xét nghiệm và thu tiền từ bệnh nhân - Ngày 18/7/2025

 Xử lý đơn XN

 Lịch sử thanh toán

 Tìm kiếm...

Lịch sử thanh toán

Tổng cộng 2 giao dịch



Nguyễn Văn An

SĐT: 0901234567

630.000 đ

Tiền mặt

Dịch vụ:

Xét nghiệm CD4 và Viral Load

550.000 đ

Xét nghiệm kháng định

80.000 đ

Thanh toán lúc:

23:31:49 18/7/2025

Trạng thái:

Đã thanh toán



Lê Minh Cường

SĐT: 0903456789

550.000 đ

Tiền mặt

Dịch vụ:

Xét nghiệm CD4 và Viral Load

550.000 đ

Thanh toán lúc:

23:28:35 18/7/2025

Trạng thái:

Đã thanh toán

- Function Details:
 - Display information of payment transaction list

7. Manager System

a. Manager Dashboard

- Function Trigger: The logged in administrator accesses the path "/manager"
- Function Description:
 - Actors: Management
 - Purpose: Displays an overview of clinic operations
 - Interface: Dashboard with statistical charts and reports
 - Data Processing: Retrieves and aggregates data from multiple sources
- Screen Layout:

Dashboard Quản lý

Quản lý tổng thể hệ thống HIV Care

Tổng người dùng
20
20 đang hoạt động

Tổng lịch hẹn
2
1 hoàn thành

Tổng doanh thu
120.000 đ
Năm 2025

Tổng dịch vụ
5
5 đang hoạt động

Người dùng

Lịch hẹn

Dịch vụ

Lịch làm việc

Doanh thu

Hệ thống

Quản lý người dùng

+ Thêm người dùng

Hiển thị 20 người dùng

Q

Tìm kiếm theo username hoặc email...

Tất cả vai trò

USERNAME	EMAIL	VAI TRÒ	TRẠNG THÁI	NGÀY TẠO	HÀNH ĐỘNG
regstaff01	N/A	Nhân viên tiếp nhận	Hoạt động	3/7/2025	Hoạt động
regstaff02	N/A	Nhân viên tiếp nhận	Hoạt động	3/7/2025	Hoạt động
qqqq	tiendon23122004@gmail.com	Bệnh nhân	Hoạt động	4/7/2025	Hoạt động
ddd	dontse183237@fpt.edu.vn	Bệnh nhân	Hoạt động	4/7/2025	Hoạt động

Quản lý tổng thể hệ thống HIV Care

Tổng người dùng
20
20 đang hoạt động

Tổng lịch hẹn
2
1 hoàn thành

Tổng doanh thu
120.000 đ
Năm 2025

Tổng dịch vụ
5
5 đang hoạt động

Người dùng

Lịch hẹn

Dịch vụ

Lịch làm việc

Doanh thu

Hệ thống

Quản lý dịch vụ

+ Thêm dịch vụ

TÊN DỊCH VỤ	LOẠI	GIÁ	MÔ TẢ	HÀNH ĐỘNG
Tư vấn	Tư vấn	100.000 đ	Tư vấn về cho bệnh nhân HIV qua online	Sửa Ẩn
Khám tổng quát HIV	Khám bệnh	150.000 đ	Khám sức khỏe tổng quát cho bệnh nhân HIV	Sửa Ẩn
Xét nghiệm CD4 và Viral Load	Xét nghiệm	550.000 đ	Đếm tế bào CD4 và đo tải lượng virus HIV	Sửa Ẩn
Xét nghiệm khẳng định	Xét nghiệm	80.000 đ	Xét nghiệm khẳng định HIV, CD4 và Viral Load	Sửa Ẩn
Xét nghiệm sàng lọc	Xét nghiệm	120.000 đ	Sàng lọc HIV sơ bộ	Sửa Ẩn

● Function Details:

- Display statistics on the number of patients by day/week/month
- Display statistics on revenue by service, by time
- Display statistics on the performance of doctors and staff

- Allow management to export reports in multiple formats

8. Guest Features

a. Home Page

- Function Trigger: User accesses home page "/"
- Function Description:
 - Actors: All users
 - Purpose: Displays general information about the clinic and services
 - Interface: Home page with banner, service introduction, contact information
 - Data Processing: Retrieves information from database
- Screen Layout:

Trung tâm chăm sóc HIV hàng đầu Việt Nam

"Đồng hành cùng sức khỏe của bạn"

Chăm sóc toàn diện, chuyên nghiệp và bảo mật

Giới thiệu trung tâm

Về chúng tôi

Với hơn 20 năm kinh nghiệm, chúng tôi cam kết mang đến dịch vụ chăm sóc sức khỏe chất lượng cao cho bệnh nhân HIV/AIDS.

Sứ mệnh

Chăm sóc toàn diện và tận tâm cho mọi bệnh nhân, góp phần xây dựng cộng đồng khỏe mạnh và không kỳ thị.

Tầm nhìn

Trở thành trung tâm chăm sóc HIV/AIDS hàng đầu khu vực, tiên phong trong công nghệ và phương pháp điều trị.

Giá trị cốt lõi

Tận tâm, chuyên nghiệp, hiện đại và luôn đặt bệnh nhân làm trung tâm trong mọi hoạt động.

Đội ngũ bác sĩ chuyên nghiệp

Các chuyên gia hàng đầu trong lĩnh vực HIV/AIDS



Đội ngũ bác sĩ chuyên nghiệp

Các chuyên gia hàng đầu trong lĩnh vực HIV/AIDS



BS. Nguyễn Minh Khoa

Xem chi tiết



BS. Trần Thị Lan

Xem chi tiết



BS. Vũ Văn Tú

Xem chi tiết

Dịch vụ nổi bật

Các dịch vụ chăm sóc sức khỏe toàn diện



Xét nghiệm CD4 và Viral Load

Đếm tế bào CD4 và đo tải lượng virus HIV

550.000đ

Xem chi tiết



Xét nghiệm sàng lọc

Sàng lọc HIV sơ bộ

120.000đ

Xem chi tiết

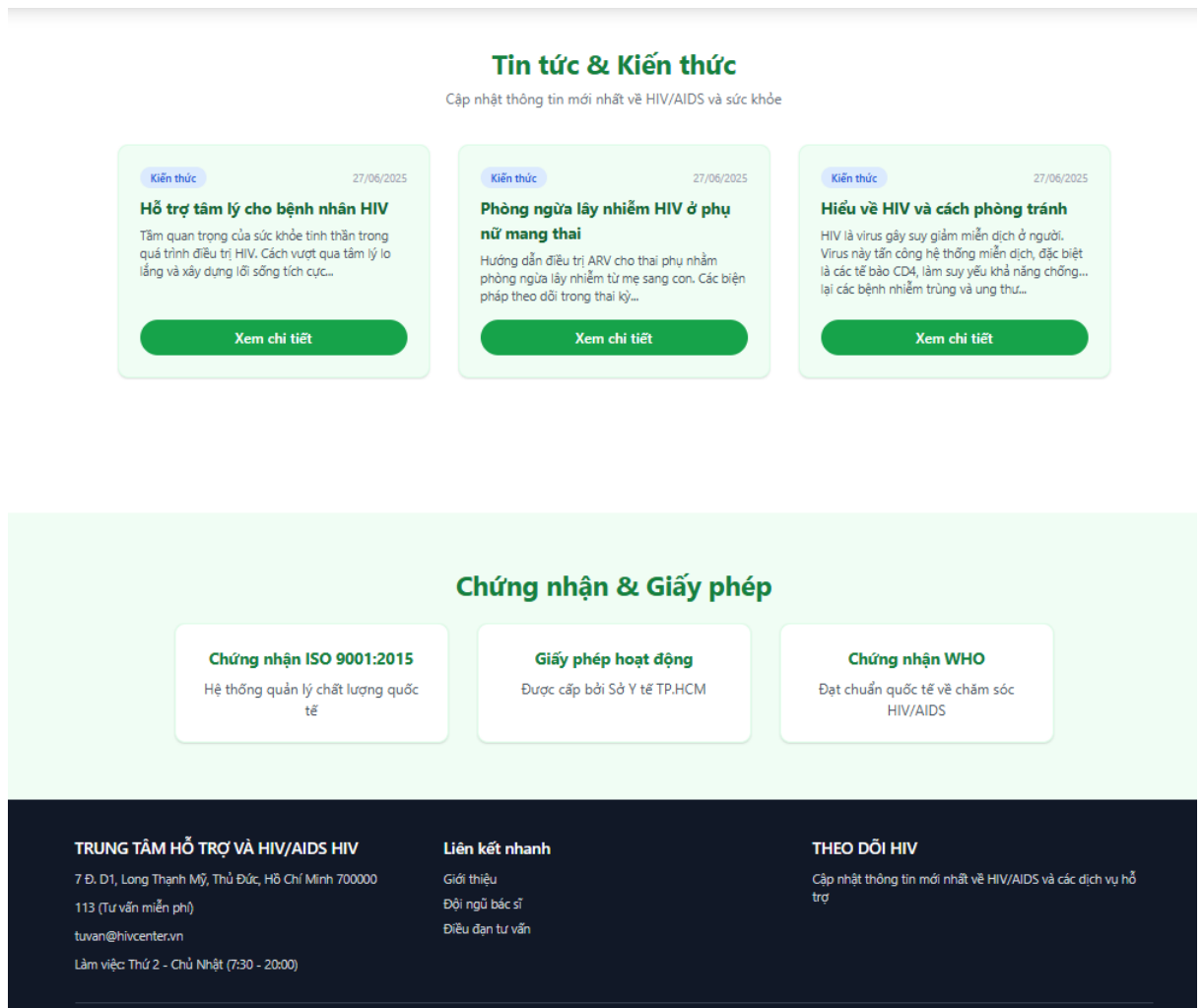


Xét nghiệm khẳng định

Xét nghiệm khẳng định HIV, CD4 và Viral Load

80.000đ

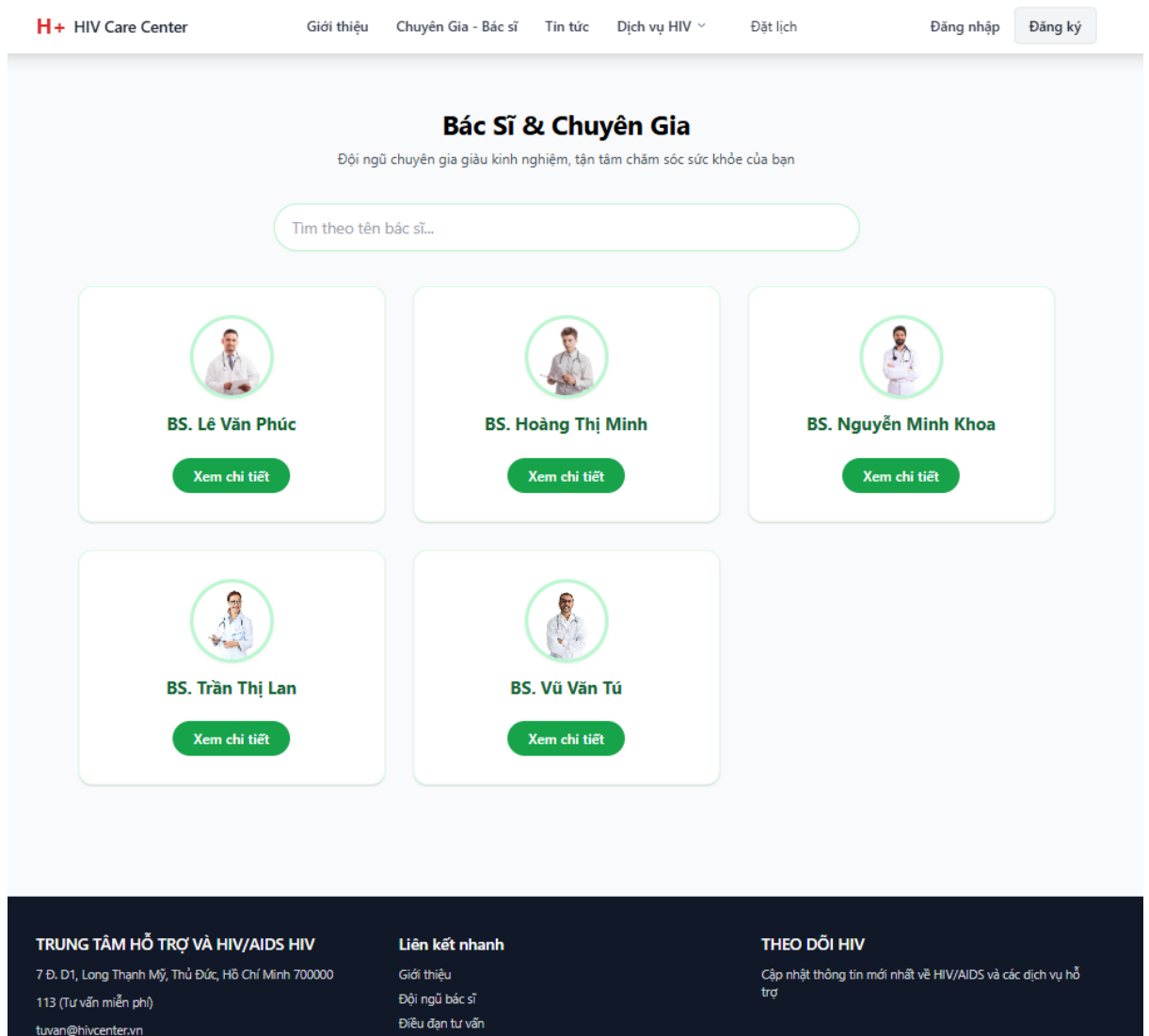
Xem chi tiết



- Function Details:
 - Display banners and key messages
 - Display a list of featured services
 - Display information about the medical team
 - Display patient reviews
 - Display contact information and clinic location

b. Doctor Directory

- Function Trigger: User accesses the path "/doctorpage"
- Function Description:
 - Actors: All users
 - Purpose: Display doctor list and details
 - Interface: List page with filter and doctor details page
 - Data Processing: Retrieve doctor information from database
- Screen Layout:



- Function Details:
 - Show list of doctors with basic information (name, specialty, experience)
 - Allow filtering of doctors by specialty, experience
 - Allow searching for doctors by name
 - Show doctor detail page when clicking on a specific doctor
 - Provide "Make appointment" button to make appointment with doctor

c. About Page

- Function Trigger: User accesses the path "/about"
- Function Description:
 - Actors: All users
 - Purpose: Display detailed information about the clinic
 - Interface: Information page with introduction, history, mission
 - Data Processing: Retrieve content from database
- Screen Layout:

Giới thiệu về HIV Care Center

Trung tâm tiên phong trong chăm sóc, xét nghiệm và điều trị HIV/AIDS tại Việt Nam

Về trang chủ



Lịch sử & Sứ mệnh

Được thành lập từ năm 2010, HIV Care Center là đơn vị tiên phong trong lĩnh vực phòng chống, xét nghiệm và điều trị HIV/AIDS tại Việt Nam. Chúng tôi cam kết mang lại dịch vụ y tế chất lượng cao, bảo mật và không kỳ thị cho mọi bệnh nhân.

- Sứ mệnh: Chăm sóc toàn diện, đồng hành cùng bệnh nhân HIV/AIDS.
- Tầm nhìn: Trở thành trung tâm hàng đầu khu vực về điều trị HIV/AIDS.
- Giá trị: Tận tâm, chuyên nghiệp, hiện đại, nhân văn.

Đội ngũ chuyên gia



BS. Lê Văn Phúc



BS. Hoàng Thị Minh



BS. Nguyễn Minh Khoa



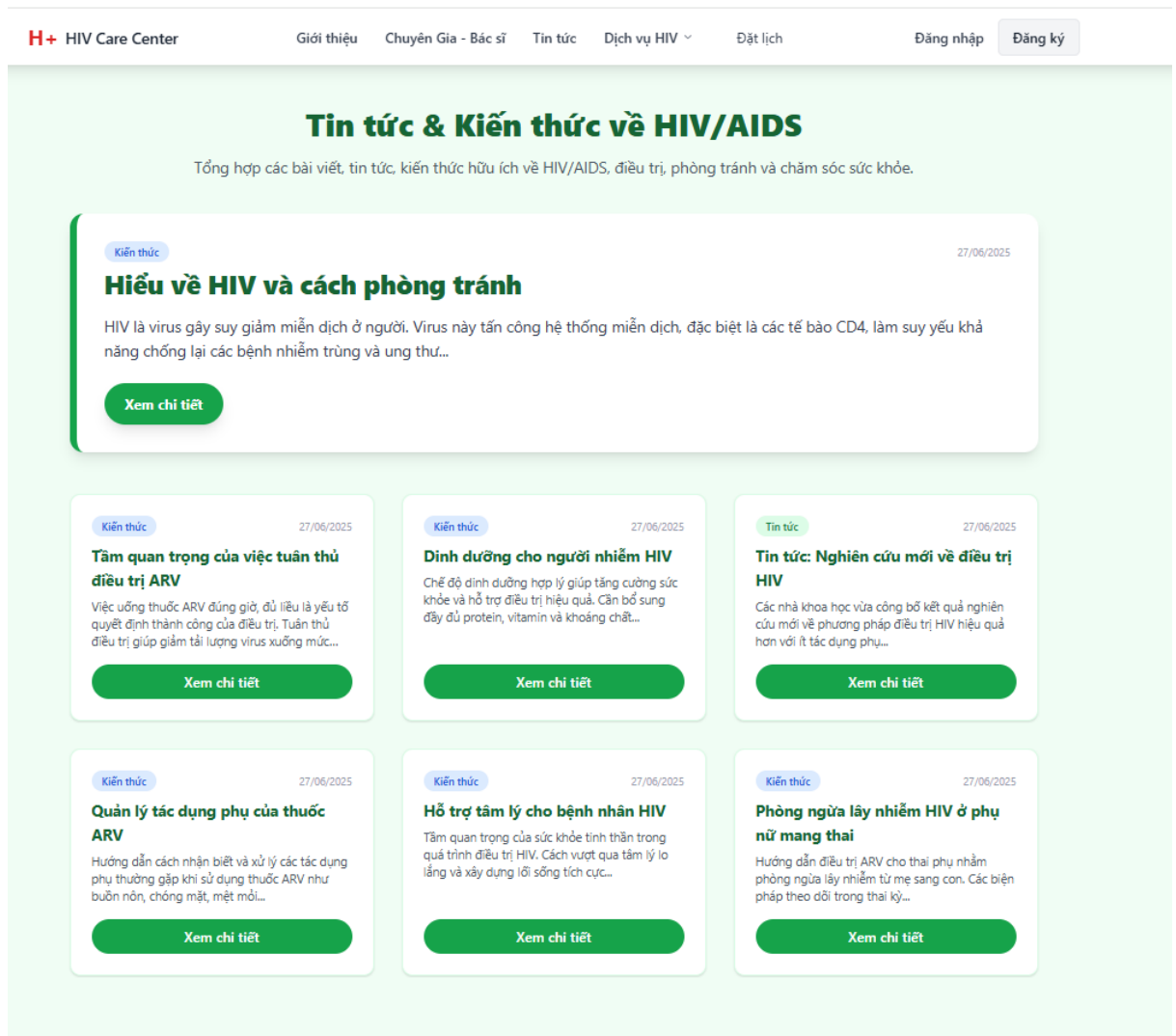
BS. Trần Thị Lan



BS. Vũ Văn Tú

d. News Page

- Function Trigger: User accesses the path "/news"
- Function Description:
 - Actors: All users
 - Purpose: Display health related news and articles
 - Interface: News page with article list and detail page
 - Data Processing: Retrieve article content from database
- Screen Layout:



- Function Details:
 - Show list of articles with title, summary, image
 - Allow to filter articles by topic, time
 - Show article detail page when clicking on a specific article
 - Allow to share articles on social networks

III. Database Design

1. Database Schema

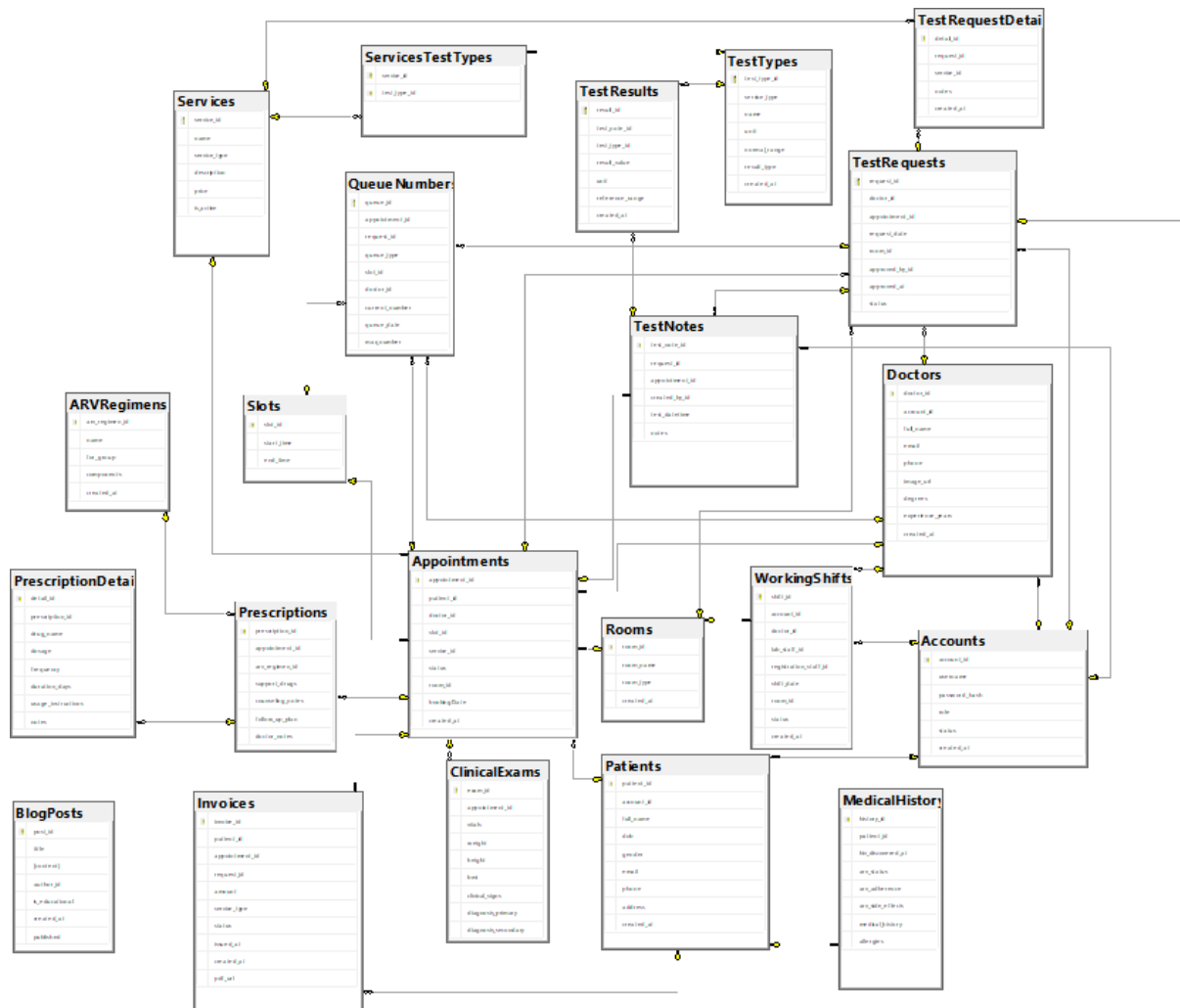


Table Description

No	Table	Description
1	Accounts	Stores user authentication details and roles (Patient, Lab-Staff, Registration-Staff, Doctor, Manager).
2	Patients	Holds personal and contact information for patients linked to their accounts.
3	Doctors	Stores details of doctors, including qualifications and contact information.
4	ARVRegimens	Contains antiretroviral regimens for HIV treatment, including components and target groups.
5	Rooms	Manages room details for appointments and test requests.
6	Slots	Defines fixed time slots for appointments throughout the day.
7	Services	Lists available services (e.g., test, examination, consultation) with prices.
8	TestTypes	Defines types of tests with units and normal ranges.
9	ServicesTestTypes	Junction table linking Services and TestTypes for test-specific services.

10	Appointments	Tracks patient appointments, including doctor, slot, service, and status.
11	TestRequests	Manages test requests issued by doctors, with approval by Registration-Staff.
12	TestRequestDetails	Stores details of services within a test request.
13	WorkingShifts	Schedules shifts for staff (Doctors, Lab-Staff, Registration-Staff) by date and room.
14	TestNotes	Records notes for test procedures, linked to requests or appointments.
15	MedicalHistory	Maintains patient medical history, including HIV and ARV details.
16	ClinicalExams	Records clinical examination details for appointments.
17	Prescriptions	Manages prescriptions issued during appointments, including ARV regimens.
18	PrescriptionDetails	Details specific drugs within a prescription.
19	BlogPosts	Stores news or educational content for the website.
20	Invoices	Tracks billing details for appointments or test requests.
21	QueueNumbers	Manages queue numbers for examinations or tests, separate from appointment logic.

2. Table Detail

Accounts

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	account_id	INT			X	PK	
2	username	NVARCHAR	50	X	X		
3	password_hash	VARCHAR	255		X		
4	role	VARCHAR	20		X		
5	status	VARCHAR	10		X		active or inactive
6	created_at	DATETIME			X		GETDATE()

Patients

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	patient_id	INT			X	PK	
2	account_id	INT		X	X	FK	
3	full_name	VARCHAR	100		X		
4	dob	DATE			X		
5	gender	VARCHAR	10				
6	email	VARCHAR	100	X			
7	phone	VARCHAR	20	X			
8	address	NVARCHAR	255				
9	created_at	DATETIME			X		

Doctors

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	doctor_id	INT			X	PK	
2	account_id	INT		X	X	FK	
3	full_name	VARCHAR	100		X		
4	email	VARCHAR	100		X		
5	phone	VARCHAR	20				
6	image_url	VARCHAR	255				
7	degrees	VARCHAR	500				
8	experience_years	INT	255				
9	created_at	DATETIME			X		

ARVRegimens

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	arv_regimen_id	INT			X	PK	
2	name	VARCHAR	100	X	X		
3	for_group	NVARCHAR	50				
4	components	NVARCHAR	500				
5	created_at	DATETIME			X		
6	is_active	BIT			X		DEFAULT 1

Rooms

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	room_id	INT			X	PK	
2	room_name	NVARCHAR	50		X	FK	
3	room_type	NVARCHAR	30		X		
4	email	VARCHAR	100		X		
5	created_at	DATETIME			X		
6	is_active	BIT			X		DEFAULT 1

Slots

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	slot_id	INT			X	PK	
2	start_time	TIME			X	FK	
3	end_time	TIME			X		
4	is_active	BIT			x		DEFAULT 1

5	create_at	DATETIME			X		DEFAULT GETDATE()
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Services

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	service_id	INT			X	PK	
2	name	NVARCHAR	100		X	FK	
3	service_type	NVARCHAR	30		X		
4	description	NVARCHAR	500				
5	price	DECIMAL	10, 2				
6	is_active	BIT					

TestTypes

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	test_type_id	INT			X	PK	
2	service_type	NVARCHAR	30		X	FK	
3	name	NVARCHAR	50	X	X		
4	unit	NVARCHAR	50				
5	normal_range	VARCHAR	100				
6	result_type	VARCHAR	20				
7	created_at	DATETIME			X		

ServicesTestTypes

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	service_id	INT				PK, FK	
2	test_type_id	INT				PK, FK	

Appointments

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	appointment_id	INT			X	PK	
2	patient_id	INT			X	FK	
3	doctor_id	INT				FK	
4	slot_id	INT				FK	
5	service_id	INT				FK	
5	status	VARCHAR	20				
7	room_id	INT		X	X	FK	
8	bookingDate	DATETIME					
9	created_at	DATETIME			X		

TestRequests

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	request_id	INT			X	PK	
2	doctor_id	INT			X	FK	
3	appointment_id	INT				FK	
4	request_date	DATETIME					
5	room_id	INT				FK	
5	approved_by_id	INT	20			FK	
7	approved_at	DATETIME					
8	status	VARCHAR	20				

TestRequestDetails

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	detail_id	INT			X	PK	
2	request_id	INT			X	FK	
3	service_id	INT			X	FK	
4	notes	NVARCHAR R	255				
5	created_at	DATETIME			X		

WorkingShifts

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	shift_id	INT			X	PK	
2	account_id	INT			X	FK	
3	lab_staff_id	INT				FK	
4	registration_staff_id	INT				FK	
5	shift_date	DATE			X		
6	room_id	INT				FK	
7	status	VARCHAR	20				
8	created_at	DATETIME					
9	is_active	BIT			X		DEFAULT 1

TestRequests

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
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1	request_id	INT			X	PK	
2	doctor_id	INT			X	FK	
3	appointment_id	INT				FK	
4	request_date	DATETIME					
5	room_id	INT				FK	
5	approved_by_id	INT	20			FK	
7	approved_at	DATETIME					
8	status	VARCHAR	20				

TestNotes

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	test_note_id	INT			X	PK	
2	request_id	INT				FK	
3	appointment_id	INT				FK	
4	created_by_id	INT		X			
5	test_datetime	DATETIME		X		FK	
5	notes	NVARCHAR R	500			FK	
6	create_at	DATETIME			X		DEFAULT GETDATE()

TestResults

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	result_id	INT			X	PK	
2	test_note_id	INT			X	FK	
3	test_type_id	INT			X	FK	
4	result_value	VARCHAR	100				
5	unit	VARCHAR	20				
6	reference_range	VARCHAR	100			FK	
7	created_at	DATETIME	500		X	FK	

MedicalHistory

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	history_id	INT			X	PK	
2	patient_id	INT			X	FK	
3	hiv_discovered_at	DATE			X	FK	
4	arv_status	VARCHAR	50				
5	arv_adherence	VARCHAR	20				
6	arv_side_effects	NVARCHAR R	500				
7	medical_history	NVARCHAR R	500				

8	allergies	NVARCHAR R	500				
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ClinicalExams

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	exam_id	INT			X	PK	
2	appointment_id	INT			X	FK	
3	vitals	INT			X	FK	
4	weight	NVARCHAR R	255				
5	height	DECIMAL	5,2				
6	bmi	DECIMAL	5,2				
7	clinical_signs	NVARCHAR R	500				
8	diagnosis_primary	NVARCHAR R	255				
9	diagnosis_secondary	NVARCHAR R	500				

Prescriptions

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	prescription_id	INT			X	PK	
2	appointment_id	INT			X	FK	
3	arv_regimen_id	INT				FK	
4	support_drugs	NVARCHAR R	255				
5	counseling_notes	NVARCHAR R	5,2				
6	follow_up_plan	NVARCHAR R	5,2				
7	doctor_notes	NVARCHAR R	500				

PrescriptionDetails

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	detail_id	INT			X	PK	
2	prescription_id	INT			X	FK	
3	drug_name	NVARCHAR R	100		X		
4	dosage	NVARCHAR R	50				
5	frequency	NVARCHAR R	100				
6	duration_days	INT	5,2				
7	usage_instructions	NVARCHAR R	255				

8	notes	NVARCHAR R	255				
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BlogPosts

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	post_id	INT			X	PK	
2	title	NVARCHAR R	255		X		
3	content	NVARCHAR R	MAX				
4	author_id	INT			X	FK	
5	created_at	DATETIME			X		
6	published	BIT			X		
7	is_active	BIT			X		DEFAULT 1

Invoices

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	invoice_id	INT			X	PK	
2	patient_id	INT			X	FK	
3	appointment_id	INT	100		X		
4	request_id	INT	50				
5	amount	DECIMAL	10,2				
6	service_type	NVARCHAR R	50				
7	service_type	VARCHAR	20				
8	issued_at	DATETIME					
9	created_at	DATETIME					
10	pdf_url	VARCHAR	255				

QueueNumbers

#	Field name	Type	Size	Unique	Not Null	PK/FK	Notes
1	queue_id	INT			X	PK	
2	appointment_id	INT				FK	
3	request_id	INT				FK	
4	queue_type	VARCHAR	20		X		
5	slot_id	INT				FK	
6	doctor_id	INT				FK	
7	current_number	INT			X		
8	queue_date	DATE			X		
9	max_number	INT			X		